1) a) Max Take off weight = 447,700kg, Max fuel capacity = 238,610 Liters

b) 90% wing Lift = 0.9 (447,700kg) = 402930kg = 187,309kg

c) 45% = Single wing lift force = 402930N/2 (9.81 m/s²) = 1976371.65N

Single wing weigth = 41000kg/2 • 20500kg (9.81 m/s²) = 201105N

6Engine-2867 dry weigth = 15,552lb = 5623kg (9.81 m/s²) = 55161.63N

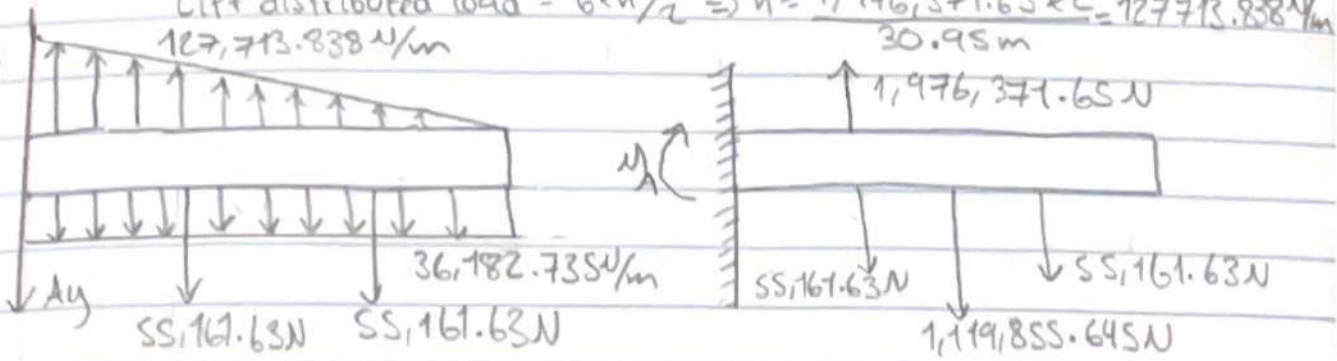
Fuel weigth = 187,309kg (9.81 m/s) = 1,837,501.29N

Fuel weigth per wing = 1,837,501.29N/2 = 918,750.645N

Fuel weigth + Wing weigth = 918,750.645N + 201105N = 1,119,855.645N

Lift distributed load = 6×h/2 ⇒ h = 1,976,371.65×2 / 30.95m = 127713.838 N/m

d)



Fuel weigth + Wing weigth = 1,119,855.645N / 30.95m = 36182.735 N/m

$\Sigma F_y = 0 \Rightarrow 1,976,371.65 - (2 \times 55161.65 + 1,119,855.645) + A_y$

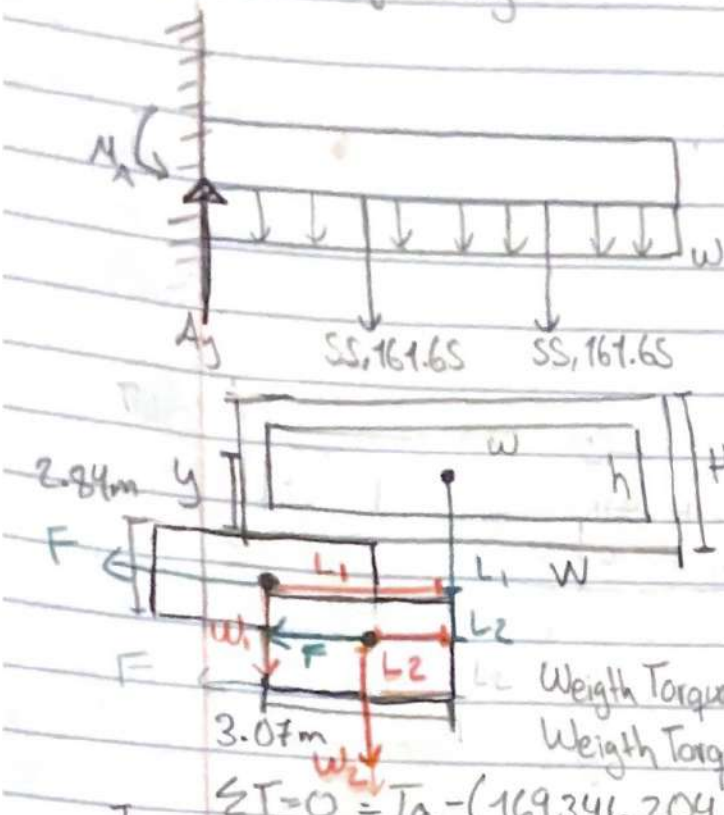
$\Rightarrow A_y = -746192.745N$

$\Sigma M_z = 0 = -1,976,371.65(10.317m) + 55,161.63(10.317m) + 1,119,855.645(15.475) + 55,161.63(20.63m) + M_A$

$\Rightarrow M_A = 1,353,373.243 N \cdot m$

Wing Distance = 61.9m/2 = 30.95m

Stationary Wing



$\Sigma F_y = A_y - (2 \times 55161.65 + 1,119,853.645) = 0$

$\Rightarrow A_y = 1230,178.905N$

$\Sigma M_2 = 0 = M_A + 55161.63(10.317m) + 1,119,853.645(15.475m) + 55161.63(20.63m)$

$w = 36182.735 N/m$

$\Rightarrow M_A = -190,36853.07 N \cdot M$

$H = 1m, \quad W = 20.78m$

$h = 0.8m, \quad w = 20.58m, \quad y = 0.5m$

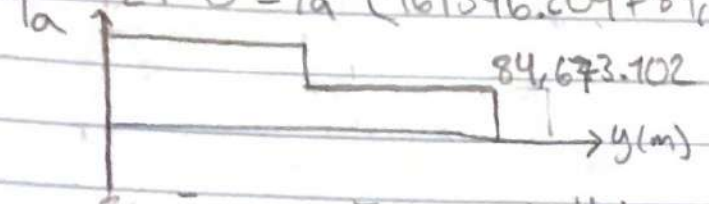
$I = \frac{WH^3 - wh^3}{12} = \frac{20.78 \times 1^3 - (20.58 \times 0.8^3)}{12}$

$I = 0.8536 m^4$

Weigth Torque 1 $= (-55161.63N)(3.07m) = -169346.204 N \cdot m$

Weigth Torque 2 $= (-55161.63N)(1.535m) = -84,673.102 N \cdot m$

$\Sigma T = 0 = T_a - (169346.204 + 84,673.102) \Rightarrow T_a = 254,019.306 N \cdot m$



$\tau_{bending} = \frac{VQ}{It}, \quad t = 0.10m$

$\tau_{torsion} = T_{max} / 2At$

$Q = \bar{A} \cdot \bar{y} \Rightarrow \bar{A} = (W-w) \cdot \frac{H-h}{2} = 20.78 - 20.58 \cdot \frac{1-0.8}{2} = 0.02m^2$

$\bar{y} = \frac{H}{2} - \frac{H-h}{4} = \frac{1}{2} - \frac{1-0.8}{4} = 0.45m \Rightarrow Q = 0.02 \times 0.45 = 0.009 m^3$

$A = (W-t)(H-t) = (20.78 - 0.10)(1 - 0.10) = 18.612 m^2$

$\tau_{torsion} = 254,019.306 / (18.612)(2)(0.10) = 68240.733 Pa$

$\tau_{bending\ with\ lift} = \frac{-76002.9N(0.02)}{0.8536(0.10)} = -17807.615 Pa = \tau_{min}$

$\tau_{nom\,a} = \sqrt{\tau_{torsion}^2 + \tau_a^2}, \quad \tau_{nom\,m} = \sqrt{\tau_{torsion} + \tau_m}$

$\tau_{bending\ stationary} = \frac{80125ON(0.02)}{0.8536(0.10)} = 187734.3018 Pa = \tau_{max}$

$\tau_a = \frac{187734.302 + 17807.615}{2} = 102,770.458 Pa$

$\tau_m = \frac{187734.302 - 17807.615}{2} = 84963.343 Pa$

The point of interest: The nominal axial stress due to bending will be more important than shear stresses due to shear forces. The point of interest could be at between y=0 to y=10.33m because is the most likely to fail because the maximum bending moment. The segment from y=0 to y=10.33m presents the maximum normal (axial) bending stress and torsion.

$M_{max} = -2232120\ N\cdot m$, $M_{min} = -190,36853.07\ N\cdot m$

$\sigma_{max} = \frac{-2232120(0.5)}{0.8536} = -1307474.227\ Pa$ $\sigma_{min} = \frac{-190,36853.07(0.5)}{0.8536}$

$\sigma_{min} = -11150921.43 \Rightarrow \sigma_a = \sigma_{max} - \sigma_{min}/2$, $\sigma_m = \sigma_{max} + \sigma_{min}/2$

$\Rightarrow \sigma_a = \frac{-1307474.227 + 11150921.43}{2} = 4.922\ MPa$

$\sigma_m = \frac{-1307474.227 - 11150921.43}{2} = -6.229\ MPa$

$r = 2.5\ m$, Al 110 sheet annealed $S_{ut} = 90\ MPa - 13\ Kpsi$ $D = 1.2\ m$

$\Rightarrow \frac{\sqrt{a}}{15-10} = \frac{0.341-0.5}{x-0.5} \Rightarrow \sqrt{a} = 0.4046$, $\frac{D}{d} = 1.2$, $D = 1.2(1)$

$q = \frac{1}{1 + \frac{0.4046}{\sqrt{2.5}}} = 0.7962$, $K_f = q(K_t - 1) + 1$ $K_t = A(\frac{r}{d})^b$

$\Rightarrow K_t = 0.99590(2.5)^{-0.23829} = 0.800$

$\Rightarrow K_f = 0.7962(0.800 - 1) + 1 = 0.8408$

$\sigma_a = K_f \sigma_{a\,nom} = 0.8408(4.922\ MPa) = 4.138\ MPa$

$\sigma_m = K_f \sigma_{m\,nom} = 0.8408(-6.229\ MPa) = -5.237\ MPa$

For torsion and shear $D = 1.20\ m$, $D/d = 1.20$, $K_{ts} = A(2.5)^b$

$\Rightarrow K_{ts} = 0.83425(2.5)^{-0.21649} = 0.6841 \Rightarrow K_{fs} = 0.7962(0.6841 - 1) + 1$

$K_{fs} = 0.7485 \Rightarrow \tau_{torsion} = \tau_m = 0.7485(68240.733\ Pa) = 51078.189\ Pa$

$\sigma'_a = \sqrt{\sigma_a^2 + \tau_a^2} = \sqrt{(4.138)^2} = 4.138\ MPa$

$\sigma'_m = \sqrt{\sigma_m^2 + 3(\tau_m)^2} = \sqrt{(-5.237)^2 + 3(51078.189)^2} = 5.238\ MPa$

$S_f' = 0.4(90\ MPa) = 36\ MPa$, $C_{load} = 1$, $C_{size} \Rightarrow A_{95} = 0.05bh$

$A_{95} = 0.05(1m)(20.78m) = 1.039\ m^2$, $d_{equiv} = \sqrt{1.039/0.0766} = 3.683\ m$

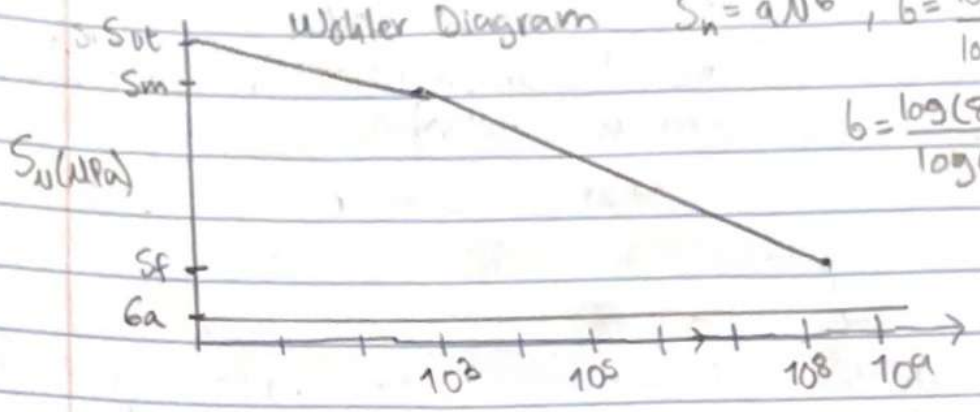
$C = 0.6$ larger size, $C_{surf} = 4.51(90\ MPa)^{-0.265} = 1.369$

Machined

C_{temp}: heat treated between 90°F and 770°F, $C_{temp} = 1$

$C_{reliab.}$ 99.9 = 0.753 $\Rightarrow S_f = (0.753)(1)(1.369)(0.6)(1)(36\,MPa)$

$S_f = 22.266\,MPa$ at 5×10^8, $S_m = 0.9(90) = 81\,MPa$



$S_n = aN^b$, $b = \frac{\log(S_m) - \log(S_f)}{\log(N_1) - \log(N_2)}$

$b = \frac{\log(81) - \log(22.266)}{\log(10^3) - \log(5\times10^8)}$

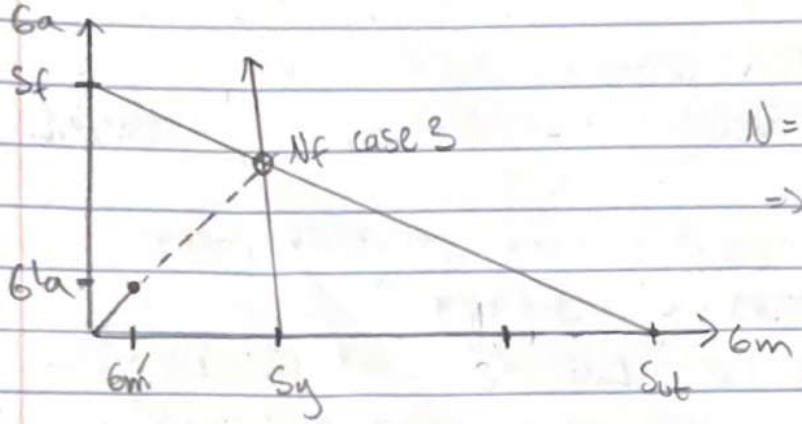
$b = -0.0984$

$a = \frac{S_n}{N^b} = \frac{S_m}{N_1^b} = \frac{81}{(10^3)^{-0.0984}} = 159.84$

$\Rightarrow S_{10^6} = 159.84(10^6)^{-0.0984}$

$S_{10^6} = 41.047\,MPa$

$S_y = 34\,MPa$



$N = \frac{S_N S_{ut}}{6'_a S_{ut} + 6'_m S_N}$

$\Rightarrow N = \frac{(41.047)(90)}{(9.138)(90) + (5.238)(41.047)}$

$N = 6.29$

14.27m

Maximum deflection when theres lift force = 3.16 mm at ~~2.02m~~

Max angle when theres lift force = ~~0.0~~196° at 2.02m

Max deflection when plane is stationary = 63.68 mm at 30.95m

Max angle when plane is stationary = 0.01836° at x = 0m